DEPARTMENT OF ARCHITECTURE AND CIVIL ENGINEERING
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Analysing Peak Loading Conditions in Office Environments

AR30315 MEng Dissertation

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Abstract

Understanding and accurately modelling extreme floor loading events in office environments is critical for ensuring safe and efficient structural designs. This study critically evaluates whether existing empirical and theoretical values, as well as modelling approaches, reflect the true characteristics of extreme loading events observed in modern offices. This was achieved through a comprehensive literature review of historical design guidance, probabilistic models, and empirical studies, with comparison to new empirical data.

It was found that current design codes typically adopt conservative, uniformly distributed imposed loads, many of which are based on outdated historical surveys. These assumptions often fail to account for the spatial and transient nature of real-world loading events. Probabilistic and statistical models aim to address this limitation through many different methods, with the most prevalent and widely accepted representing the extraordinary load event as a spike process that is later combined with the sustained load which is modelled as a Poisson arriving square. However, even these models often rely on the same outdated survey data for validation, raising questions of their relevance to present day offices which are drastically different to those at the time of the surveys.

To be able to critically analyse the findings of the literature review an original survey was distributed to office workers gathering data on respondent demographics, peak population densities during specific event types (e.g. meetings, evacuations), and short-term storage loading. The key findings were:

- Historical survey data and design recommendations frequently underrepresent modern extreme event population densities.
- Short term storage loading demonstrated a higher maximum weight per unit area than those found in previous surveys relating to population densities.
- Peak loading events were highly localised.
- Stochastic models of extreme loading may oversimplify real-world scenarios.

The study concludes by making several recommendations for future research to improve the understanding of extreme events aiming to reduce overdesign without compromising safety.

Acknowledgements

The author would like to sincerely thank Dr Will Hawkins for his invaluable support and guidance throughout this project. From helping to shape the structure of the dissertation to assisting with survey distribution and data representation, his input was instrumental to the final outcome.

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1.0 Introduction

1.1 What is the Problem?

The accurate structural assessment of office buildings heavily relies on floor loading specifications to ensure safety and performance. Extreme load cases, such as evacuations, and other intermittent loading, although not very long-lasting, often represent scenarios where maximum loading occurs and therefore set the values that are advised to be used in building design. However, as Costa and Beck (2024) state “Unlike the sustained load, survey data for the extraordinary part of the loading is notably scarce” with much of the literature focusing on extreme loading using data that is not robustly evidenced to inform modelling techniques. While these theoretical models provide a baseline for design, they lack validation through empirical data derived from real-world observations. This absence of empirical validation raises questions about the accuracy of these findings and their applicability to design. The reliance on theoretical predictions without supporting real-world evidence creates uncertainty, potentially leading to designs that either overestimate or underestimate the actual loadings experienced in office environments.

1.2 Why is it a Problem?

The discrepancy between theoretical predictions and real-world loading scenarios has far-reaching implications for structural design. One of the most significant consequences is the trend toward overdesigning structural members, which leads to inefficiencies in material usage. Overdesign not only increases construction costs but also contributes significantly to the unnecessary usage of finite materials. The construction industry is a major contributor to global carbon emissions, accounting for approximately 40% of emissions causing climate change (International Energy Agency and United Nations Environment Programme, 2018). Figure 1 shows how embodied carbon changes with imposed loading for a four-story reinforced concrete building, highlighting how even minor improvements in specified loading could result in substantial reductions in global emissions especially if implemented on an industry-wide scale.

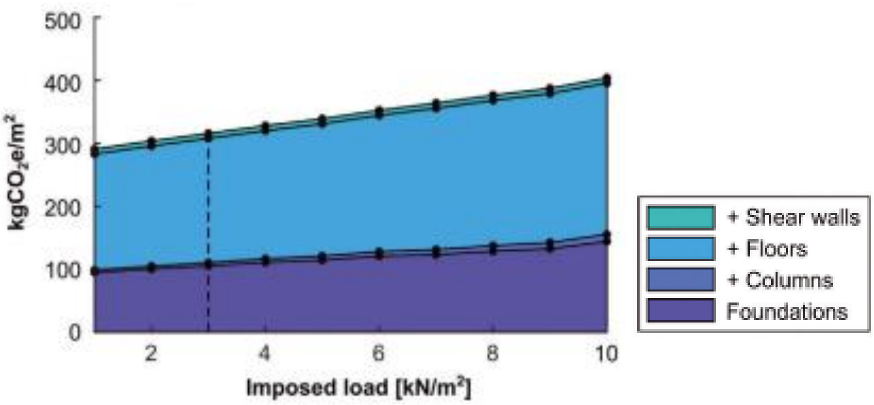


Figure 1: Variation of embodied carbon with imposed loading (Hawkins et al., 2021)

Additionally, overdesign imposes unnecessary financial burdens on construction projects. Materials typically account for 50% of a project's overall cost (Bridgit, 2024), meaning that excessive material usage directly inflates budgets. For investors and developers, these additional costs represent lost opportunities for resource allocation to other project areas.

However, reducing overdesign is not without challenges. Stricter designs aimed at minimising material use may inadvertently compromise structural safety. By reducing redundancies within the structure, these designs leave less room to accommodate unforeseen load conditions, increasing the risk of failure. Moreover, designs tailored for specific use cases, such as office environments, may lack the flexibility to adapt to changes in building function. For instance, a floor designed for maximum loads in an office setting might not accommodate the demands of a gym. This lack of adaptability could necessitate the construction of new spaces rather than repurposing existing ones, leading to additional environmental and financial costs.

1.3 Aims and Objectives

The overarching aim of this dissertation is to bridge the gap between theoretical and empirical assessments of extreme floor loadings in office environments, thereby contributing to more accurate estimations of imposed loading, and therefore sustainable, cost-effective, and adaptable design practices. The specific objectives of this research are:

1. To compare existing literature regarding extreme loading and see what the most prevalent assumptions and modelling techniques are.
2. To collect data on maximum floor loading events in office environments that is representative of modern office usage.
3. To identify trends in the data and gain insight into extreme loading conditions experienced in real-world situations.
4. To distil survey data into quantifiable parameters that can validate values provided in literature, highlighting discrepancies and proposing adjustments to design practices and theoretical assumptions where necessary.

1.4 Scope

This research focuses exclusively on office environments and does not extend to other building types, such as residential or industrial spaces. Additionally, the research is limited to comparing empirical data with findings from literature addressing maximum load cases. Direct comparisons with specific design codes or standards fall outside the scope of this study. However, the insights gained from this research may serve as a foundation for more detailed analysis and potential revisions to design practices in the future.

2.0 Literature Review

2.1 Literature Review introduction

This literature review explores the key methods used to model and design for extreme loading, specifically regarding office buildings. It explores three primary methods of research: historical and code-based design, probabilistic and statistical models, and empirical surveys. With an ever-increasing drive for sustainable efficient designs, the review aims to assess whether traditional assumptions remain valid and how closely theoretical models reflect real-world conditions. As this study focuses on the comparison of new data to existing data, the literature review forms essential context for the subsequent analysis and interpretation of results.

2.2 Historical and code-based design approaches

Traditional approaches to extreme floor loading design have primarily relied upon prescriptive values derived from historical empirical studies, rule of thumb safety margins and previous failure events. Despite significant changes in building use and occupancy patterns, many of these values exist to some extent in current design codes. A key example is the British Standard BS 6399-1 (British Standards Institution, 1996) and its successor Eurocode EN 1991-1-1(European Committee for Standardization, 2002), which have served as foundational documents for structural design in the UK and Europe over the past few decades. Both documents recommended imposed load values in the range of 1.4 to 3.0 kN/m² for office environments. These values are applied uniformly across all floor areas regardless of their function, layout or actual occupancy. Although these values are not regarding extreme events, they highlight the broad categorisation of areas in design, which this study aims to verify as overconservative by providing updated data on occupancy loading patterns across different office environments, allowing for more tailored and potentially optimised structural design approaches.

Alexander (2002) critically reevaluates the basis of such design values. A key example is a study conducted by Mitchell and Woodgate whereby they derived the 95%, 99, and 99.9% probable loadings directly from survey data. Alexander points out the difficulty in interpreting Mitchell and Woodgates results, as their presented values reflected the probability of a given load occurring on one floor at a specific moment, rather than the probability of such loading occurring at any point during the building's lifetime. As highlighted in Table 1 the values that Mitchell and Woodgate proposed, had a maximum value of 2.43kN/m². However, the load specified in row 15 is an additional mobile load which is meant to represent the movement of high loaded items or personnel, which moves the maximum load up to 3.05 kN/m². Alexander questions the justification for this and the fact that the survey data has already captured the extreme events shows that layering additional load on top may result in unnecessary conservatism. Mitchell and Woodgate's work also assumed a value of 2.4 kN/m² for crowd loading during a fire event, a value that was widely adopted by others in the field, has no derivation or explanation where the value came from. This study aims to verify these values with no reasoning behind them and see if the converted values used are accurate.

Table 1: Occupancy loading by floor area (Alexander, 2002)

1 Mean loaded area A (m^2)	5.2	14	31	58	111	192
2 Square root area √A (m)	2.3	3.7	5.6	7.6	10.5	13.9
Occupancy loading on upper floors (kN/m²)						
3 Mean loading	0.64	0.62	0.61	0.59	0.58	0.57
4 Standard deviation	0.53	0.43	0.35	0.30	0.26	0.22
5 91.9% probable loading	1.50	1.30	1.15	1.03	0.95	0.88
6 Effective standard deviation	0.61	0.49	0.38	0.32	0.26	0.22
7 Additional mobile loading	0.62	0.29	0.14	0.10	0.10	0.05
8 Total 91.9% occupancy loading	2.12	1.59	1.29	1.13	1.05	0.92
9 Load concentration factor	1.4	1.4	1.4	1.4	1.4	1.4
10 Characteristic occupancy loading	*2.96*	*2.22*	*1.81*	*1.58*	*1.47*	*1.29*
Occupancy loading on floors at and below ground level (kN/m²)						
11 Mean loading	0.68	0.67	0.67	0.65	0.64	0.62
12 Standard deviation	0.64	0.55	0.50	0.44	0.34	0.36
13 91.9% probable loading	1.81	1.58	1.50	1.35	1.16	1.19
14 Effective standard deviation	0.81	0.65	0.60	0.50	0.37	0.41
15 Additional mobile loading	0.62	0.29	0.14	0.10	0.10	0.05
16 Total 91.9% occupancy loading	2.43	1.87	1.64	1.44	1.25	1.24
17 Load concentration factor	1.4	1.4	1.4	1.4	1.4	1.4
18 Characteristic occupancy loading	*3.41*	*2.61*	*2.30*	*2.02*	*1.75*	*1.73*

In Alexander's (2002) study he also analyses a key piece of research 'An assessment of the imposed loading needs for current commercial office buildings in Great Britain,' known as the Stanhope study, released in 1992. Alexander compared the values presented against those proposed in his paper, giving the ratio in the last column of Table 2. Some values were as low as 0.22 highlighting the huge discrepancy between models and the need for verification with real values. The Stanhope study also introduced additional design recommendations stating that 5% of any floor should be designed for higher loading, with the rest designed at a base value of 2.5 kN/m². Although this wasn't widely adopted by industry, the BCO (British Council for Offices) specifications stated that this 5% value should be as high as 7.5 kN/m². This is one of the earliest values for an extreme localised load and acted as a precedent for future studies, although Alexander believes these values to be overconservative.

Table 2: Comparison with the Stanhope study (Alexander, 2020)

Description	Dimensions	Area (incl partitions)	Occupancy loading	Partition loading	Imposed loading	Proposed value in paper	Ratio
Library A	4.10 x 3.20 m	13.86 m²	24.0 kN	5.3 kN	2.11 kN/m²	3.25 kN/m²	0.65
Filing room	5.60 x 3.55 m	20.80 m²	51.7 kN	6.5 kN	2.80 kN/m²	2.98 kN/m²	0.94
Stanhope office	6.50 x 5.90 m	39.60 m²	20.2 kN	8.8 kN	0.73 kN/m²	2.64 kN/m²	0.28
Financial advisors' office	5.75 x 5.75 m	34.22 m²	12.0 kN	8.2 kN	0.59 kN/m²	2.71 kN/m²	0.22
Library B	7.00 x 3.50 m	25.56 m²	24.4 kN	7.5 kN	1.25 kN/m²	2.86 kN/m²	0.44
Presentation room	4.00 x 4.80 m	20.09 m²	26.3 kN	6.3 kN	1.62 kN/m²	3.00 kN/m²	0.54
Packed room	4.25 x 2.80 m	12.40 m²	28.9 kN	3.5 kN	2.61 kN/m²	3.34 kN/m²	0.78
Dealing floor	4.20 x 8.00 m	34.83 m²	28.2 kN	8.7 kN	1.06 kN/m²	2.70 kN/m²	0.39

The need for empirical grounding is taken further by Hawkins et al. (2021). They provide further critique of these codes through their study on the environmental consequences of over design due to overspecification of load assumptions. Much of their analysis uses data gathered through the Minimising Energy in Construction (MEICON) initiative. They argue that reducing imposed loads could yield material savings of 4% /m^2. They demonstrate that real loads in buildings are substantially lower than design codes values, with the mean-imposed load in reality being 0.6kN/m². They suggest that 99.7% of floors had loading below 2.5kN/m². They also use the MEICON data to show that the average specified floor load is 3.1kN/m2 which is represented in Figure 2. This shows an average overspecification of 24% contributing to a huge unnecessary material use. Although their study does not focus on the extreme events, it provides valuable insight into the average specified load and the drastic effect load overestimation has, especially regarding the environment and building re use. This study aims to compare real data to those being used in industry to see if these values of overspecification seem accurate.

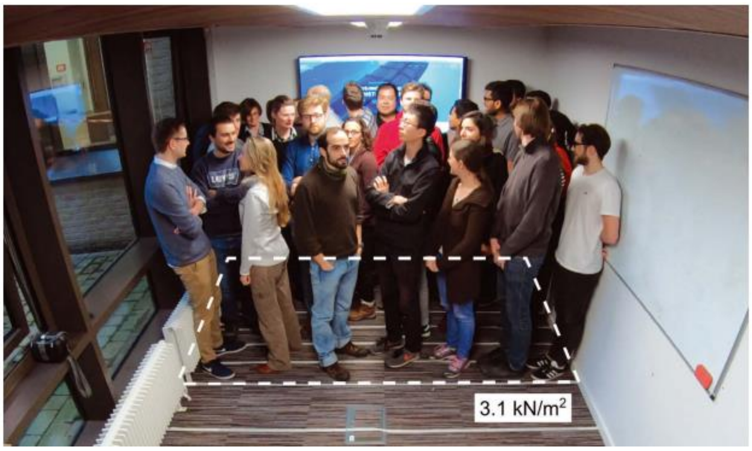


Figure 2: Average load used in office design, 3.1kN/m, represented as a crowd with over four people per square meter (Hawkins, et al., 2021)

A key piece of work by Orr (2019) echoes the concern that these codes lack empirical data to support them and warns that gradual reductions in safety factors are being introduced 'without reference to what happens in the real-world.' He stresses the need for up-to-date measurements, stating 'A key missing link in reducing uncertainty in design is data […] what loading really occurs?' showing the industry-wide need for more current data and the importance of verification before design choices are altered. This sentiment directly supports the motivation for this dissertation.

Historic work conducted by the Building Code Committee (BCC, 1925) highlights how early classification of rooms and buildings led to blanket load requirements which were extremely inefficient. Although classification was common in design, the BCC thought that the limited number of classes and the fact all areas in the same class were being designed for the heaviest possible loading seen by any in that class, was uneconomical. The report also specifies that spaces where crowds of people are likely to occur such as corridors and lobbies must be designed for a uniform load of 100 lbs/ft² (4.8kN/m²). However, a separate survey that the BCC reviewed showed a maximum loading of only 40.2 lbs/ft² (1.9kN/m²), highlighting the huge difference between experienced and recommended values. This mismatch between design assumptions and actual loadings reinforces the long-standing tendency toward overdesign in structural codes, which this study aims to minimize through its findings improving the understanding of the loads that often govern design.

The English Heritage (1996) report, highlighting the work done by the Building Research Station (BRS), discusses crowd densities in extreme events. It shows that a load of 2.4kN/m² is reached at the point when people are packed so tightly that movement is restricted to shuffling. They believe that in situations of panic with people running this density is not possible and the spacing of people would increase, reducing the static load. However, they show that the dynamic forces would have greater effect. Despite the complexity, the report believes a design value of 2.5kN/m² remains an adequate assumption This again highlights how static design values often serve as conservative approximations, not precise representations of real conditions.

Allen (1958) analyses the Canadian National Building Code, showing designs must be able to support a uniformly distributed live load of 50lb/ft (2.4kN/m²) or a concentrated load of 2000lb (907kg) distributed over an area of 25 by 25ft (58m²), to represent the extreme loading, which as a smaller uniformly distributed load is the equivalent of (15.6kN/m²). He highlights how crowding up to 180lb/ft² (8.6kN/m²) has been proved possible but only in elevators, and therefore that floor load specification is usually only 100lb/ft² (4.8kN/m²). This presents the idea that loads could be underspecified by the design codes as more extreme localised loading is possible. He stresses the importance of categorising storage by magnitudes instead of item type and that descriptive terms for each room under consideration should be used over the general term office. These terms should be based on magnitude and arrangement of person, property or storage stating, 'the variability of extreme values will depend greatly on how well the "use" is defined'. Allen's view that descriptive, use-based classifications can improve design accuracy agrees strongly with many of the views already expressed and this study aims to verify these views with empirical data to show the variation of extreme events by room type.

However, Allen (1958) presents a study that found the maximum load per unit area due to people was nearly independent of floor area and that the dependency on floor area arose from the furniture. Almost all design codes factor into their calculations a reduction factor for larger size rooms, and therefore data is needed to verify whether it is accurate that the maximum density of people is independent of room size

Collectively, these studies expose a consistent issue: traditional loading assumptions, while historically grounded, are increasingly misaligned with modern usage patterns and environmental priorities. Overdesign persists due to conservative modelling, vague classifications, and a lack of recent field data. The present study contributes to addressing this gap by gathering new empirical data on peak loading conditions in varied office environments. In doing so, it offers an evidence-based perspective on whether imposed load values can be safely reduced, with implications for both design efficiency and structural safety.

2.3 Statistical and Probabilistic Load Models

Traditional methods of extreme loading analysis relied on fixed values derived from empirical studies often with over conservative safety factors. These models lacked the ability to capture the high variability and transient nature of real loading events. This has driven the development of statistical and probabilistic load models, particularly attempting to better reflect extreme short duration events. This section traces the evolution of such models, compares their key assumptions and methods, and aims to identify shortcomings that my dissertation can address or validate.

Horne (1951, cited in Pier and Cornell ,1973. and Costa and Beck 2024) is widely regarded as the first to start theoretical probabilistic modelling of floors. His approach was simplistic, assuming that load intensities at two locations were independent of each other and time. He assumed the load had a normal distribution with a standard deviation inversely proportional to the square root of the area. While useful as a starting point and acting as a basis for many of his predecessors' work, it failed to account for the interconnected nature of loads. Rosenbluth (1959, cited in Costa and Beck, 2024) expanded on this through the introduction of influence surfaces, this allowed live loads to relate to the effects they caused on structural members. Costa and Beck (2024) show that an influence surface is effectively a graph whose ordinates show the variation of the magnitude of a structural response that is produced by a unit load applied at the point where the ordinance is measured. The development of these early probabilistic models represented a shift from purely empirical rule of thumb values to frameworks that would hopefully more accurately represent load cases.

Notably a study by Hasofer (1968) proposed that total live load on building floors could be modelled by a two-dimensional stationary stochastic process with independent increments (i.e. a compound Poisson process indexed on the loaded area A instead of time). He suggests that the tail of the distribution of total live load could be well approximated by the Pareto formula. However, this analysis method is disputed by other authors in the field, highlighting the discrepancy between the methods used by researchers, and the need for more emperical data to verify the extreme values that currently have little validation

The next critical advancement contributing to extreme loading events came from Peir and Cornell (1973), who were among the first to use the term 'transient live load' described as a load that 'happens somewhat infrequently, but with a relatively high intensity and short duration'. They identified that these extreme events were often caused by crowds of people which occurred often due to parties or 'open houses'. Their proposed models incorporated both spatial and temporal variability. They had realised that office loading fluctuates in a manner not well captured by the fixed design values previously used. They witnessed that these short-term high intensity loads caused a spike which was a stark contrast to the sustained load. Thus comparing these events to other transient events such as high wind loads or earthquakes. They developed a two-component model that distinguished between sustained loads Q(t) and extraordinary loads P(t). They proposed a stochastic representation where:

- Q(t) is modelled as a Poisson square wave process
- P(t) is modelled as a Poisson spike process
- L(t) = Q(t) + P(t) captures the combined process

Graphical representation of these models can be seen in Figure 3, and these assumptions underpin almost all further research in this field.

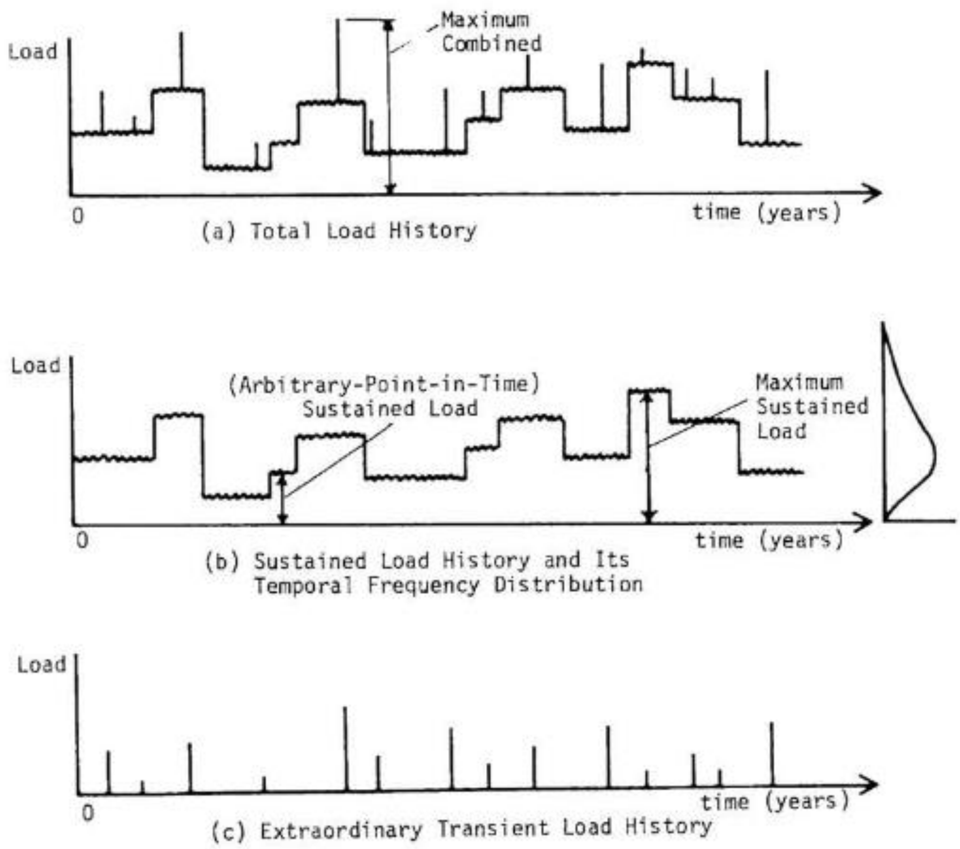


Figure 3: Graphs of total load history, sustained load history and its frequency distribution, and extraordinary transient load history (Pier and Cornell, 1973)

A crucial advancement in probabilistic modelling came from the introduction of the linear stochastic models that allowed arbitrary point in time values to map extreme events. Andam (1986) shows that these models represent the load intensity $P_{ij}(u, v)$ acting on a floor as a random function represented by:

$$P_{ij}(u, v) = V + \varepsilon(u, v)$$

V is the spatial average and $\varepsilon(u, v)$ is a stochastic process with zero mean. By integrating over a designated influence area and applying appropriate weighting, through influence functions, $P_{ij}(u, v)$ can be converted to a value called the Equivalent Uniformly Distributed Load (EUDL) by dividing by the total influence area. Pier and Cornell (1973) show that the EUDL provides a useful means to interpret load surveys and to estimate model parameters. In their paper they use the EUDL for many different analyses and comparison.

Andam (1986) believes the model to evaluate the extraordinary load E proposed by McGuire and Cornell (1974) remains one of the most reliable. Each extraordinary load is represented by Poisson arriving events, the clustering of loads cells in a room of influence area A is spatially controlled, and the number of loads per cell R. W is the weight of a single load, usually that of one person. Therefore, the moment associated with E are:

$$E[E] = \frac{\lambda \mu_W \mu_R}{A}$$

$$V_{ar}[E] = \frac{\lambda(\mu_R \sigma_W^2 + \mu_W^2 \sigma_R^2 + \mu_W^2 \mu_R^2)}{A^2}$$

The influence area used in extraordinary load modelling is scaled using a parameter K, which accounts for the clustering of live loads around structural elements. A value of K=2.2 which corresponds to column load is widely accepted.

The terms $\mu_R \sigma_R^2$ are the mean and variance respectively of the load per cell. $\mu_W \sigma_W^2$ are the mean and variance respectively of the weight of a single load. λ is a function of A representing the average number of loads cells in A.

To calculate the maximum extraordinary load, the distribution of the maximum effect of a random number of Poisson arriving loads can be shown as:

$$f_E(x) = \exp\left(-v_e T[1 - F_E(x)]\right)$$

T can be first approximated as the duration of the maximum sustained load, In Andam's study however he takes it as the design lifetime which for his analysis was taken as 64 years. $F_E(x)$ *is* the cumulative distribution which in McGuire and Cornell (1974) is a gamma distribution. v_e is the mean arrival rate of extraordinary loads, as Andam (1986) found the average occupancy time to be 15.7 years in his study $v_e T$ was 4.1.

These values can then be combined with the maximum sustained load which would have been analysed to find the 0.9 and 0.99 fractals to gather the design load. Andam's (1986) paper runs through the calculations with survey results and comes to a design load for a typical clerical room of 20 m^2 to be 1500N/m2 which is comparably less than the values shown in historical and code-based design. Figure 4 shows the variation of load effects with influence area the bottom most line represents the simulation model for the extraordinary load only.

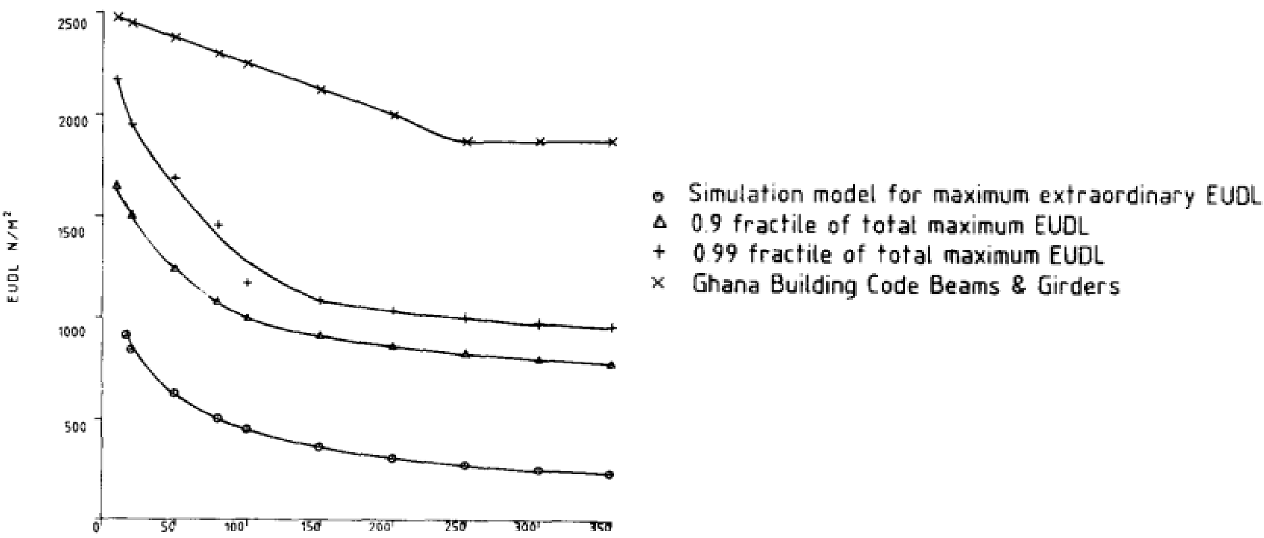


Figure 4:Fractiles of load effect and influence area (Andam, 1986)

Costa and Beck (2024) consider the work done by the Joint Committee on Structural Safety (JCSS, 2001) who believe the extraordinary load can be regarded as deterministic or as a random variable and that the duration of the pulses is almost always smaller than the pulse intervals and therefore shows a stochastic process. In the model, if multiple extraordinary loads are considered, each is represented with individual stochastic process with independent characteristics.

Although not contributing to the advancement of the model presented, a simple study by De Sanctis, et al. (2019), provides a key example of where real data has been used to verify load modelling approaches. The study gathered empirical extreme occupancy data in a supermarket environment. The authors proposed plotting the results against gamma and logarithmic distributions as shown in Figure 5, ultimately, they found better alignment with gamma models across most retail types. Although their study focused on retail environments and fire safety, the use of fitted statistical models to represent short term peak occupancy provides a valuable precedent which is equally relevant in office environments. Their study supports that extreme loading should not be treated as uniform or fixed values but rather occurrences with distributions that need mapping. The gamma and logarithmic graphs although simplistic, serve as an underpinning for many of the foundational assumptions of more in depth analysis. However, their study presented no focus on the variability of loads by room type or area which is central to this dissertation.

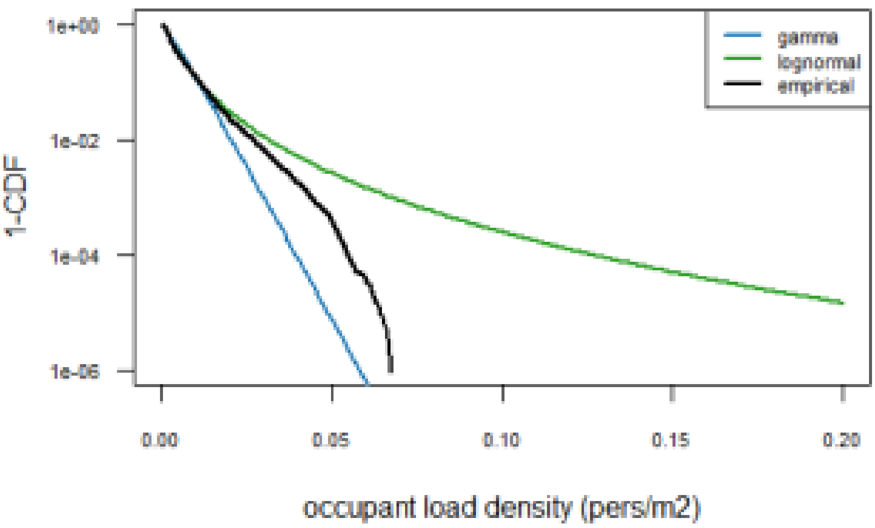


Figure 5: Example data fitted to gamma and lognormal distributions (De Sanctis, et al., 2019)

Although Peir and Cornell's work was published in 1973 little additional work has focused on the extreme loading events with only a few papers advancing this technique. However, the next key addition to this research comes from Costa et al. (2022) who have published several papers in recent years regarding this extreme load analysis almost 50 years later. They built directly on this foundation with their critical review of these and other existing live load models. They formalised this dual component approach into a hierarchical stochastic model. It models magnitudes of loads with a lognormal or gamma distribution, interval durations with exponential distribution, and the amount with a Poisson distribution. This can be seen graphically in Figure 6 and gives a more in depth and digestible representation of the assumptions and parameters of the model propose by Pier and Cornell (1973).

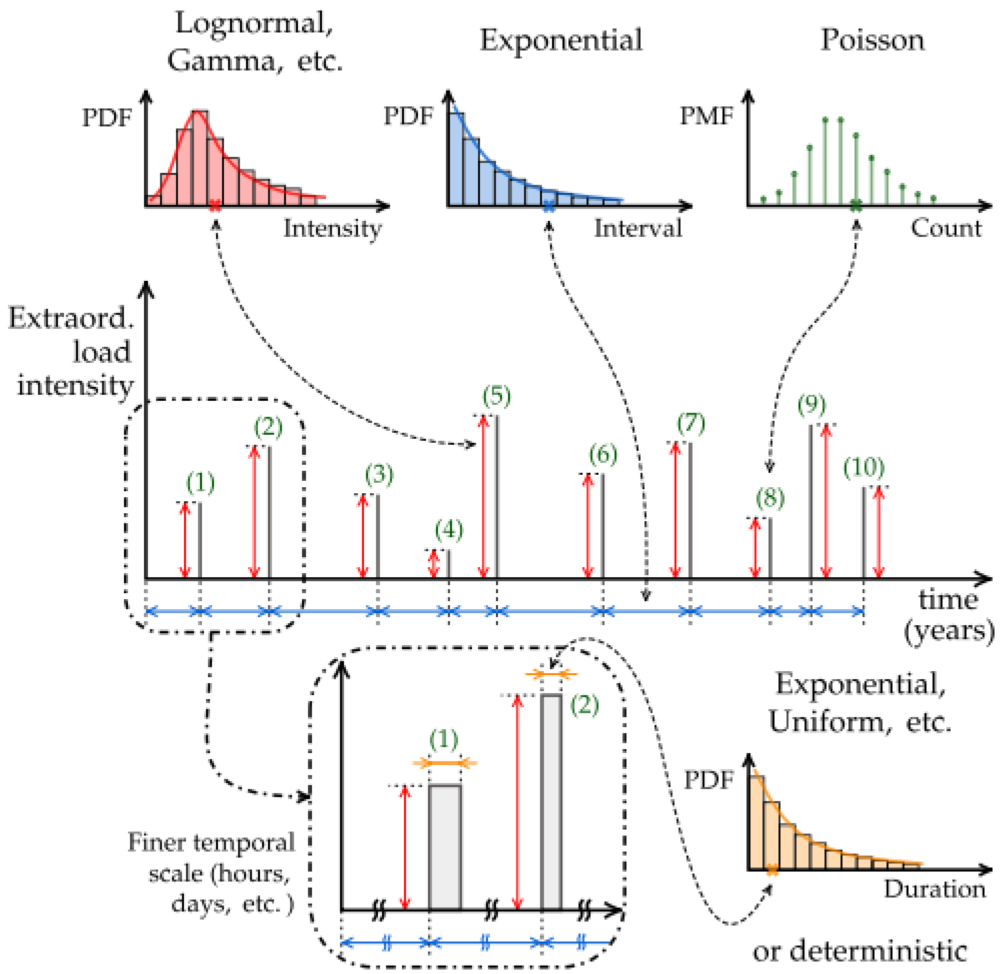


Figure 6: Time history of the extraordinary load (Costa and Beck, 2024)

However, unlike older models, Costa and Beck (2024) clearly visualises the combination, and evaluates the effect of spikes overlapping. The method of combination of the sustained and extraordinary loads has been widely disputed, however Costa and Beck do so using a Monte Carlo simulation. This is effectively a mathematical technique that simulates possible results for an event using randomly evolving values in a range rather than set values. This method allows for many random values to be generated in an aim to model more accurately.

Which type of distribution to use to model these events is highly disputed and many authers have found different ones mapping better to their values depending on the scheme of their study. The JCSS (2001) prescribe the use of an exponential distrubution to be used for the arbitary point in time load. However others argue for the use of a gamma distrubution as this is what many other previous studies use in ther analysis.

While these probabilistic models seem mathematically robust, their accuracy depends on the reliability of input data. The limitations of stochastic models were highlighted by Doggett (2022), who states ‘a significant issue […] was the inadequacy of the stochastic models which in turn is due to a limited number of surveys and empirical data representing the actual loads in buildings. The existing surveys are limited and do not take all aspects into account such as the duration of extraordinary loads'. This critique reinforces the necessity of updated empirical data to calibrate or validate the assumptions in these models.

2.4 Empirical Studies of Imposed Loading

In the previous sections empirical studies have been proven to be a vital source of insight to validate the assumptions that are embedded in live load models and design codes. While theoretical and statistical frameworks offer the ability to simulate variability and extreme events, they are limited without reliable real-world data to calibrate against. Historically, survey-based investigations have been used to quantify imposed floor loads across a range of space types, though these studies often differ in duration, methodology, and the conditions under which data was gathered.

One early example of a survey techniques was conducted by Dunham et al. (1952). Their research focused on the extreme loading events such as classrooms being intentionally overloaded. The classroom which would normally have one teacher and 48 students, was loaded with 258 students, by double seating students on chairs and filling all aisles and open spaces. This caused a load of 41.7 lbs/ft² (2.0kN/m²). From Table 3 it can be seen that the value of the maximum loads experienced was 78.3 lbs/ft (3. 7kN/m²). This rudimentary method of extreme loading analysis, just fitting as many people as you can in a space, highlights how little understanding there is about extreme events, and even the data from studies may not be representable of real conditions. Their study carries on to talk about loads coming from the storage of items, as they realise that these events are often the source of high concentration loads. They present a design requirement of 2000lb (8.9kN), coming from the weight of a safe, to be designed for on 'a limited area.' This highlights the huge variability in specification as this limited area is open to interpretation. This study aims to show what an accurate extreme load is and its accompanying area distribution.

Table 3: Maximum, minimum, and average live loads in office buildings (Dunham, et al.,1952)

	Offices	Maximum	Minimum	Average
		Lbs./ft²	Lbs./ft²	Lbs./ft²
Light-occupancy floor (twentieth)	67	55.4	0.87	10.26
Medium-occupancy floor (thirty seventh)	64	30.73	3.27	10.67
Heavy-occupancy floor (eleventh)	62	33.84	5.0	13.96
Total and average	193	-----	----	11.6
Selected heavy occupancies throughout the building	14	$\boxed{78.3}$	21.4	42.4

Another notable example of extreme load modelling is presented by De Sanctis et al. (2019), who investigated population densities in a supermarket environment to inform safe egress strategies during fire events. Their approach originally estimated occupant numbers using receipt counts, but relied on key assumptions, particularly the length of stay of the customers and the number of people per receipt. This led to them switching to light sensors to gather data allowing them to develop the model shown in Figure 7, which incorporates the 'dwell time' a value to account for time spent in the store. The study revealed significant variability in population densities, especially during sales periods, where crowding intensified dramatically. A snippet of the measured values can be seen in Table 4, which indicates a maximum witnessed density in the range of 0.35-0.4 people/m^2. Although their analysis was retail focused and surveyed extreme loading of whole floors rather than in localised areas, their methodology to model short duration peak events is relevant to office buildings where similar crowding events occur.

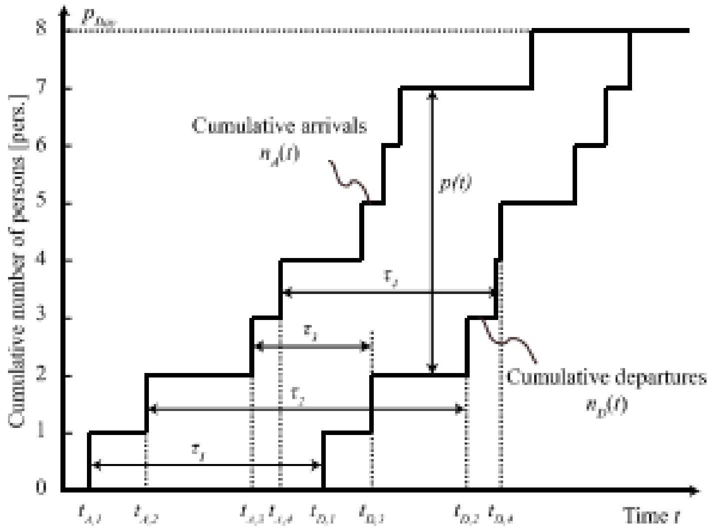


Figure 7: Cumulative arrivals and departures from a system showing the parameter 'dwell time' (De Sanctis, et al., 2019)

Table 4: Number of stores per retail type and 99% quantile value of the occupant load density (De Sanctis, et al., 2019)

Retail type	99% values of the occupant load density [pers./m^2]									
	0 - 0.05	0.05 - 0.1	0.1 - 0.15	0.15 - 0.2	0.2 - 0.25	0.25 - 0.3	0.3 - 0.35	0.35 - 0.4	0.4 - 0.45	0.45 - 0.5
Furnitures	6	0	0	0	0	0	0	0	0	0
Hardware (DIY)	6	0	0	0	0	0	0	0	0	0
Sports	5	0	0	0	0	0	0	0	0	0
Clothing	6	0	0	0	0	0	0	0	0	0
Electronics	2	8	0	0	0	0	0	0	0	0
Supermarket_600	0	7	2	0	0	0	0	0	0	0
Supermarket_1200	0	6	1	0	0	0	0	0	0	0
Supermarket_2400	1	7	0	0	1	0	0	0	0	0
Supermarket_4800	0	10	0	0	0	0	0	0	0	0
Supermarket_large	3	3	1	0	0	0	0	0	0	0
Department store	1	1	1	0	0	0	0	0	0	0
Mall	9	2	2	0	0	0	0	0	0	0
Highly frequented	0	0	0	1	0	3	0	1	0	0

Another large-scale survey was conducted by Culver, as cited in Alexander (2002), who collected information on 1354 rooms across 23 buildings. Alexander believed that due to the study concentrating primarily on the use of the space, he could not use their values to validate his study's findings. He argued that, as there is little ability to predict the use of a structure at design stage, specifying specific loads for different areas in a building meant that this limited what these spaces could be used for. This highlights how more localised analysis of structures may lead to less ability for new building re use in the future. However, Hawkins et al. (2021) show how lower load specification can lead to an increase in the ability of vertical extensions and reuse of foundations of existing buildings. This is due to the existing buildings effective overspecification allowing for additional load to be added. As construction shifts from new build to building re-use, the positive implications of more rigorous load specification seem to outweigh the negative effects.

A thesis written by Doggett (2022) regarding measurement systems for estimating imposed loads, includes a brief discussion of the early work conducted by Lars Sentler. Sentler's 1975 report reviewed various load models and drew comparisons on them, highlighting their advantages and disadvantages. Sentler identified two distinct branches of imposed loading:

- the applied - relating to data collected from surveys, measurements, and observed load patterns in buildings.
- the theoretical - focusing on conceptual understanding and the use of modelling approaches for representing these loads in design.

Sentler's distinction between applied and theoretical approaches is especially important as it highlights not only the importance of the empirical data that this study aims to gather, but also that the frameworks and models used in different cases need to accurately map situations. Sentler emphasised the need for further research in both areas, particularly in the collection of reliable empirical data.

Costa et al. (2024) reinforce the significance of empirical data to inform modelling, stating that 'the progress in the probabilistic modelling of live loads has been closely linked to the realization of various load surveys […] which play a crucial role in providing the necessary data for the development and application of analytical models.' However, they then go on to highlight limitations of such studies. Most commonly that the type of study done is one in which a survey is conducted at one point in time and then for analysis the load intensity is assumed to be an ergodic stochastic process. This effectively means that in a survey of different rooms each with point in time values, it is assumed each individual survey represents a point in time on a line as if it was one room being surveyed. A graphical representation of which is shown in Figure 8. Although this allows for the data collection at a single point in time to be used to infer the behaviour of the room over time, it is unknown how accurately this maps into reality and Costa et al. (2024) warn that this approach has a lack of physical verification. This criticism supports the need for this study to capture the high intensity loading to provide empirically grounded data to validate the assumptions.

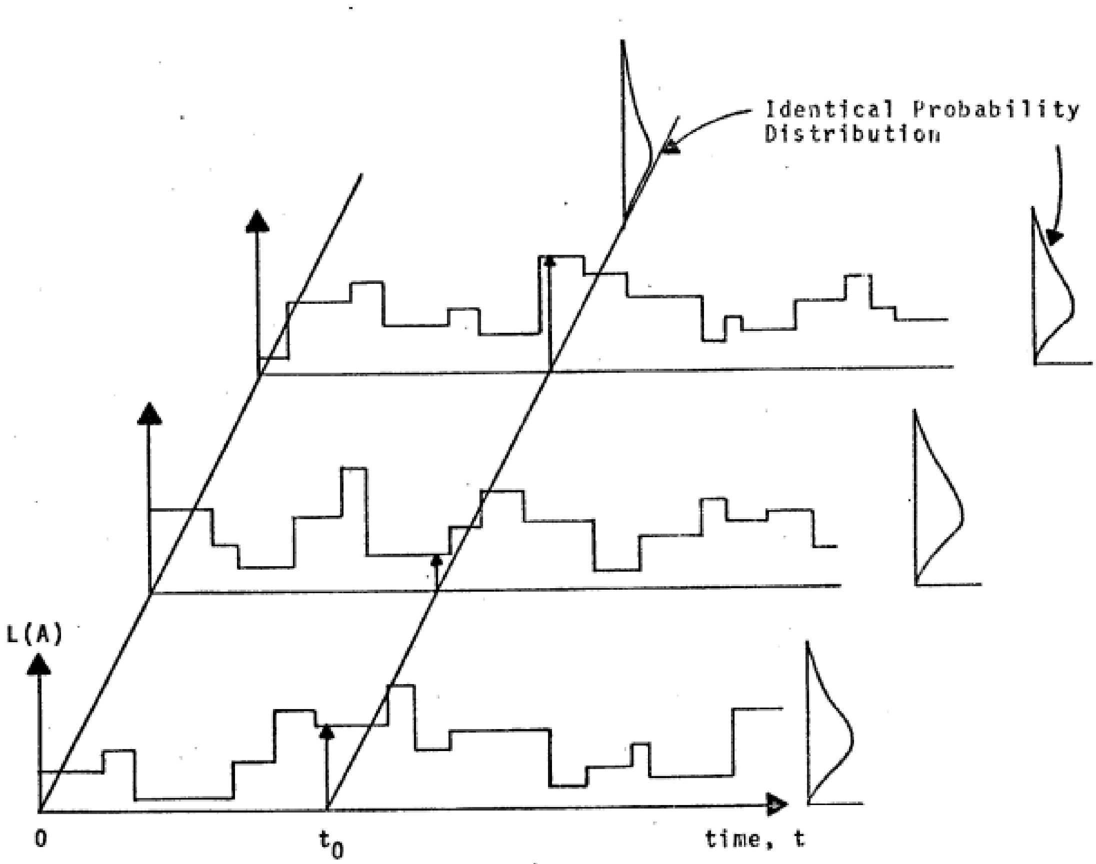


Figure 8: Three Different Load Histories and their Temporal and Spatial Frequency Distribution showing the ergodic stochastic mapping process (Peir, 1972)

As floor loading is a global problem, Andam's (1986) study in Accra, the capital city of Ghana in Africa, that compares his results to previous surveys conducted in the UK and USA is extremely useful. Surveying 1353 rooms, Andam aimed to combine empirical data with probabilistic modelling to formulate values that would facilitate computational analysis of loading events, taking into account both sustained loads and extraordinary load events calculated with an 'acceptable level of risk'. From the survey data presented in Table 5 it can be calculated that the mean personnel load for all rooms was found to be 74kN/m². Additionally, this figure shows that corridors are often significantly loaded. However, the study's preliminary research showed that kitchens were unlikely to experience large loads and were therefore not measured in detail. The current study aims to see if the omission of certain spaces is justified and if trends identified regarding extreme loads in different room types is still found in current data.

Andam (1986) shows how during a structures lifetime it is characterised by the overloading events it witnesses, typically; parties, open days, furniture stacking and crowding during an emergency. The current study aims to gather data on all event types and identify whether some important cases have been overlooked.

Table 5: Table showing measured personnel loads by room use (Andam, 1986)

Room use	Personnel load N/m^2	* Sustained load (N/m^2)					
		Maximum	Minimum	Skewness	Kurtosis	Standard deviation	Mean
Clerical	77	2500	40	3.5	24.1	200	324
General	109	2500	60	3.7	16.7	395	336
Corridor	36	640	16	1.3	1.5	144	207
Reception	24	490	20	2.1	6.2	93	112
Conference	31	770	90	1.6	4.2	137	259
Storage	40	3190	70	2.6	9.5	586	743
File	70	2000	190	1.3	2.5	412	725
Library	36	1170	160	0.2	0.3	245	646
All rooms	74	3190	16	3.8	26.7	250	334

* Includes personnel loads.

Despite the diversity of these studies, a clear pattern emerges. Existing empirical research often suffers from insufficient sample size, lack of temporal understanding of loads, or a narrow contextual focus. Additionally, many of the large surveys conducted were done over 30 years ago, meaning that their results are unlikely to map a modern office environment

This dissertation directly addresses that gap by providing updated empirical data from a wide range of office environments, specifically targeting short-duration, high-occupancy scenarios often missed by broader surveys. It seeks not only to validate or challenge existing load assumptions but also to inform future design guidance with context-specific data that better reflects how office spaces are actually used.

3.0 Methodology

3.1 Introduction

This methodology section outlines how the aims and objectives of the dissertation have been addressed through the selected research methods. Data has been gathered in two primary ways:

1. A literature review of global publications related to extreme floor loading. This includes outdated or narrowly focused studies with similar aims to this study.
2. An online questionnaire distributed primarily via LinkedIn and shared directly with individuals likely to have relevant insights or experiences of office loading. (For full questionnaire see Appendix A)

The literature review was essential to highlight the lack of real-world data and the heavy reliance on theoretical assumptions. In contrast, the questionnaire was necessary to gather practical, real-world data, allowing for a direct comparison and assessment of the accuracy of existing assumptions.

3.2 Research Philosophy and Approach

The survey used a mixture of qualitative and quantitative answers. This allowed for numerical analysis and pattern recognition while still giving the flexibility to capture all events witnessed (George, 2025). The survey was posted to LinkedIn and other social media platforms to try and gain responses from as wide a background as possible, rather than relying on engineering connections. A survey-based approach was essential, as it enabled the collection of real-world data to compare against theoretical values and assumptions presented in the literature. Additionally, respondents were asked to answer regarding their current office only. This was to ensure consistency in responses, avoid retrospective bias and improve the accuracy of the data.

The survey received approval from the University of Bath's ethics review process prior to distribution. Participation was entirely voluntary, and all responses were anonymous. Participants were informed of the nature and purpose of the questionnaire and study, and gave informed consent before completing the survey. No identifying information was gathered, and all data was stored securely in accordance with ethical research practices.

3.3 Survey Design

The questionnaire was split into three main sections:

1. Questions regarding respondent demographics
2. Questions regarding extreme event population data
3. Questions regarding extreme short term storage loading

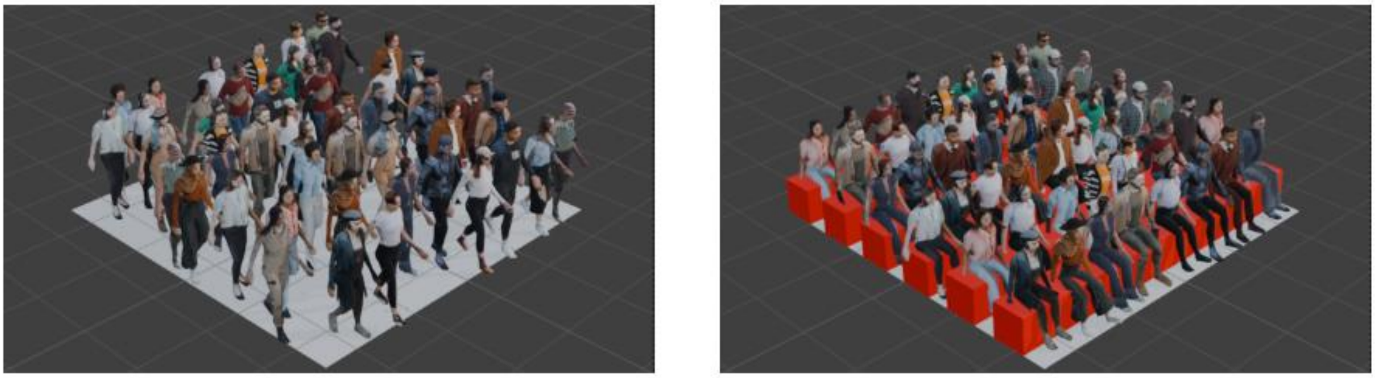
Section 1 had two questions:

1. What type of work does your office undertake?
2. How long have you been working in your current office?

Both questions used drop-down option boxes. These were implemented across the survey as they encourage participation by making questions seem less daunting and the survey quicker to complete (Sopact University, 2024).

Section 2 was presented in a table format with the following rows showing the choices of extreme events; Meeting, Party, Practice evacuation, Real evacuation, Talks/Seminars, and Other. This meant that respondents were not limited to the typical options and could choose an event that may have otherwise been overlooked. The headings of the table were: Maximum density witnessed, How many people were present at this density, How many people were present in total, How often does this type of event occur, What type of room has this event occurred in. Respondents had the ability to choose multiple different room types for the same event so that events likely to occur in several locations could be mapped effectively.

As the survey was designed to be easily accessible to individuals with no background in structural engineering or floor loading, reference images illustrating different population densities in both sitting and standing conditions, an example of which is shown in Figure 9, were used. This meant that no prior knowledge was needed to be able to visualize the loading densities. Additionally, each event type was clearly defined and described (for descriptions see Appendix A), reducing the likelihood of confusion among participants.



(2 People per m^2)

Figure 9: Visual representation of occupant density at 2 people/m². Images created using Blender to illustrate typical spatial configurations for standing (left) and seated (right) arrangements. Used in survey to aid participants' understanding.

Section 3 was formatted similarly to Section 2 with a table format and drop-down menus. The column headings were as follows, Please describe the item(s) being stored, Number of items being stored, How were they stored, Approximate floor area occupied by the stored objects, How often does this storage events occur. The table had multiple rows so that more than one event could be answered for. The survey deliberately did not get respondents to give direct weights of objects or sizes, as a lay person typically struggles to report these values accurately especially due to the size weight illusion (Wolf and Drewing, 2020). Therefore, weights are assumed from the information on the item instead. A list of example commonly stored items was given to aid recollection of events that may have otherwise been missed.

3.4 Analysis approach

The results of the survey were then analysed and formatted in a way that allowed comparison to the literature presented. This involved a key conversion of people/m^2 to kN/ m^2 as this is the units most often used. Additionally, to allow distinct data points to be plotted, many graphs used values that convert the range chosen to its median value. This approach allowed the identification of trends, anomalies, and the comparison with literature.

4.0 Results

This section presents the findings of the questionnaire, based on 25 meaningful responses.

4.1 Respondent Demographics:

Figure 10 shows the proportion of respondents by type of work. The data reflects a range of different office-based roles across many sectors, including administrative, technical and service based positions. This figure gives an overview of the background of the participants and helps to establish context on the types of office loadings that will be represented in the results.

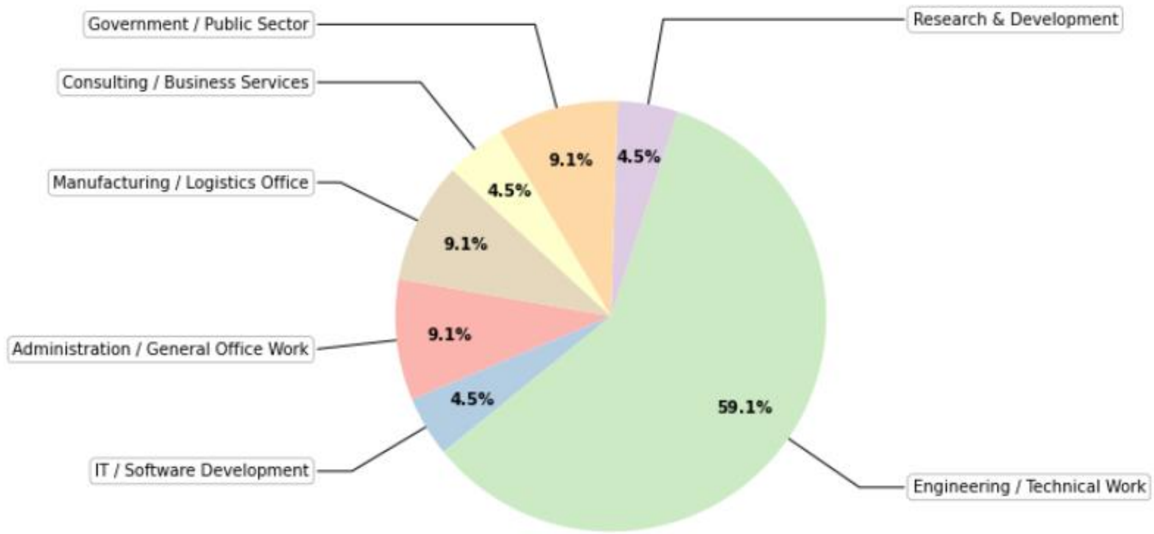


Figure 10: Proportion of respondents by type of work

Figure 11 shows the proportion of survey respondents by the time spent working in their current office. Responses covered almost the full range of responses available with results coming from those who had been working there for 4-6 months and those for 16-20 years.

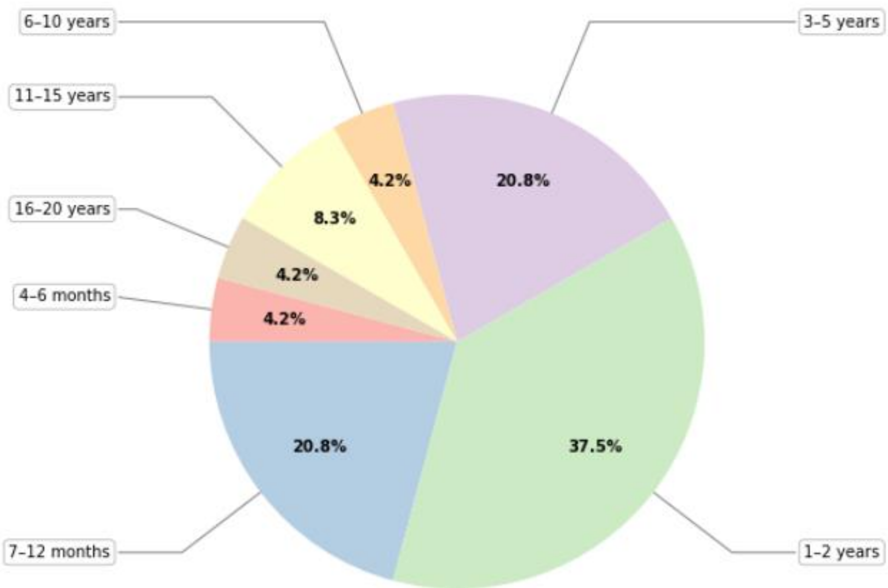


Figure 11: Proportion of respondents by time at current office.

4.2 Event based data:

Figure 12 shows the maximum density witnessed in people/m^2 by event type. Values shown are medians of the selected ranges, and the spread of results has been represented through box plots with median values represented by the thick black line and outliers by dots.

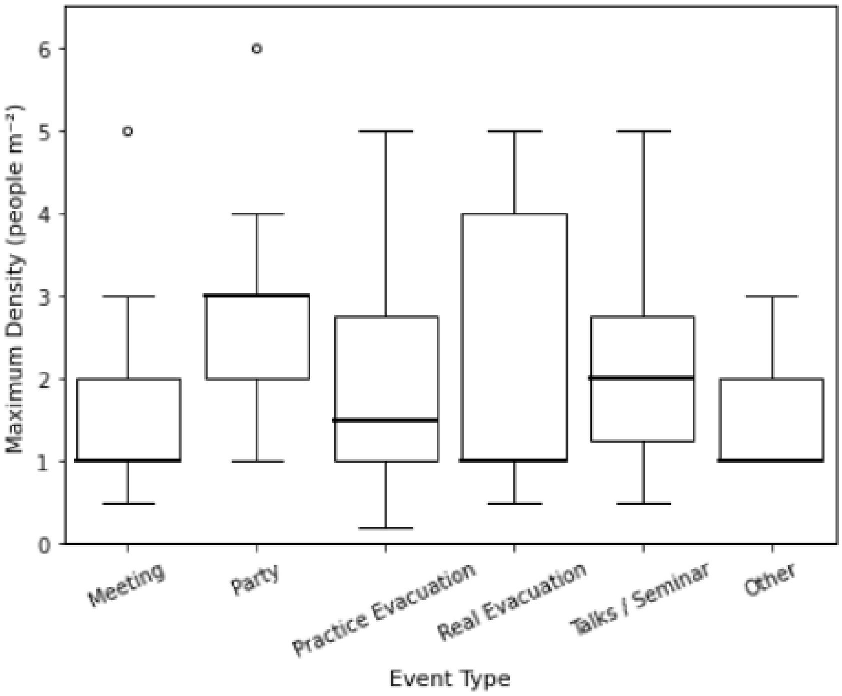


Figure 12: Reported maximum densities by event type.

Figure 13 presents the maximum density plotted against the room type. The same method of ranges being converted to median values has been used.

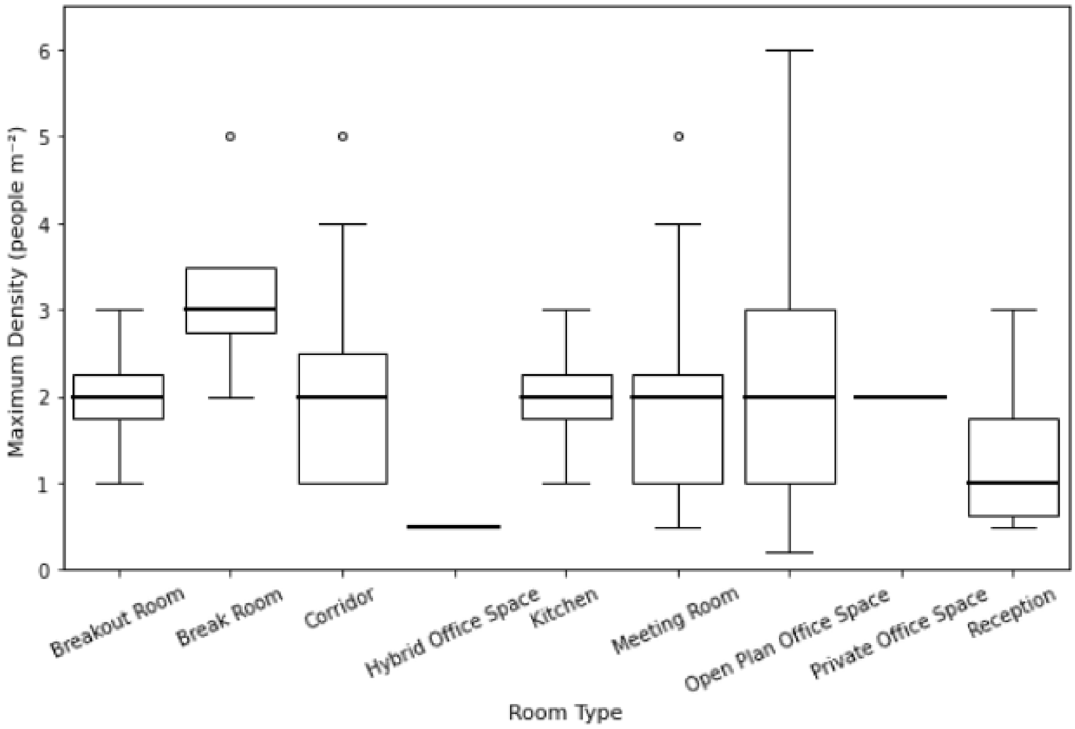


Figure 13: Reported maximum densities by room type.

Figure 14 presents the distribution of reported attendance across different event types. The orange boxplots represent the total number of people present at each event type, while the blue boxplots indicate the number of people present at the maximum density.

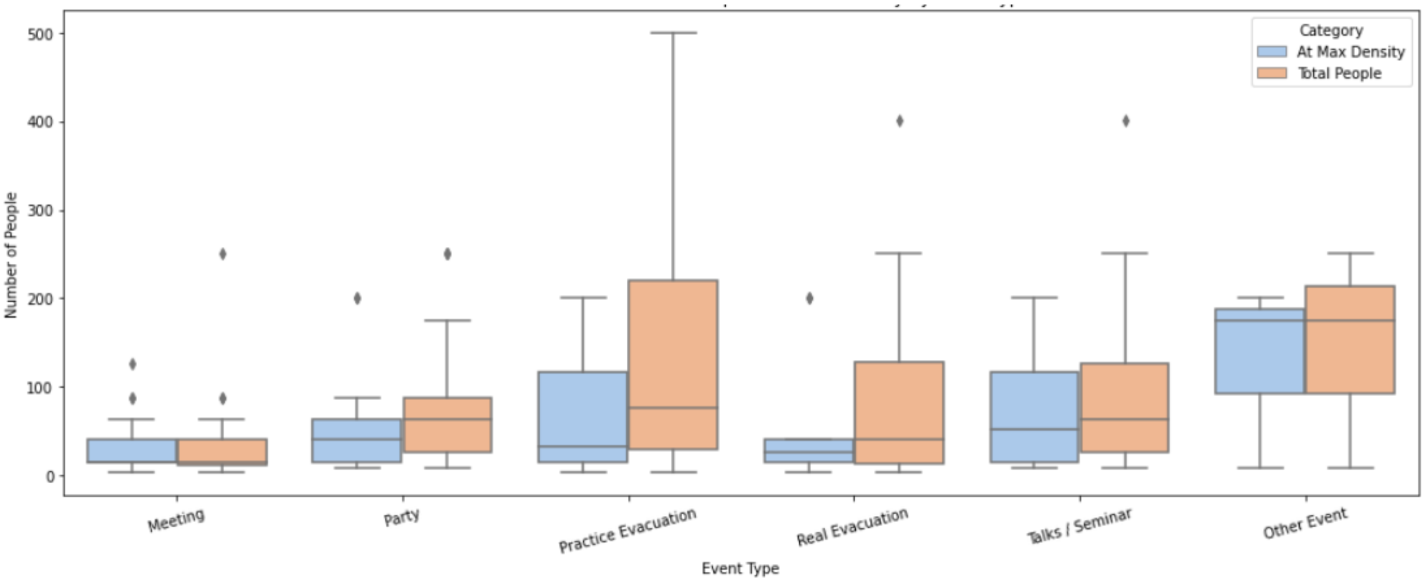


Figure 14: Reported number of people present in total and at maximum density, by event type.

Figure 15 shows the total number of responses received for each event type and the relative percentage of that event happening in each room type. The total number of responses for each event is noted in parentheses bellow the corresponding event. Respondents were able to choose multiple rooms for each event, and therefore the total value of responses for meetings (28) being above that of total participants (25) is not an error. If respondents chose multiple room types, they were asked to rank them in the order of likelihood that the event would occur, however this was done poorly and was unfortunately unable to be integrated into the results.

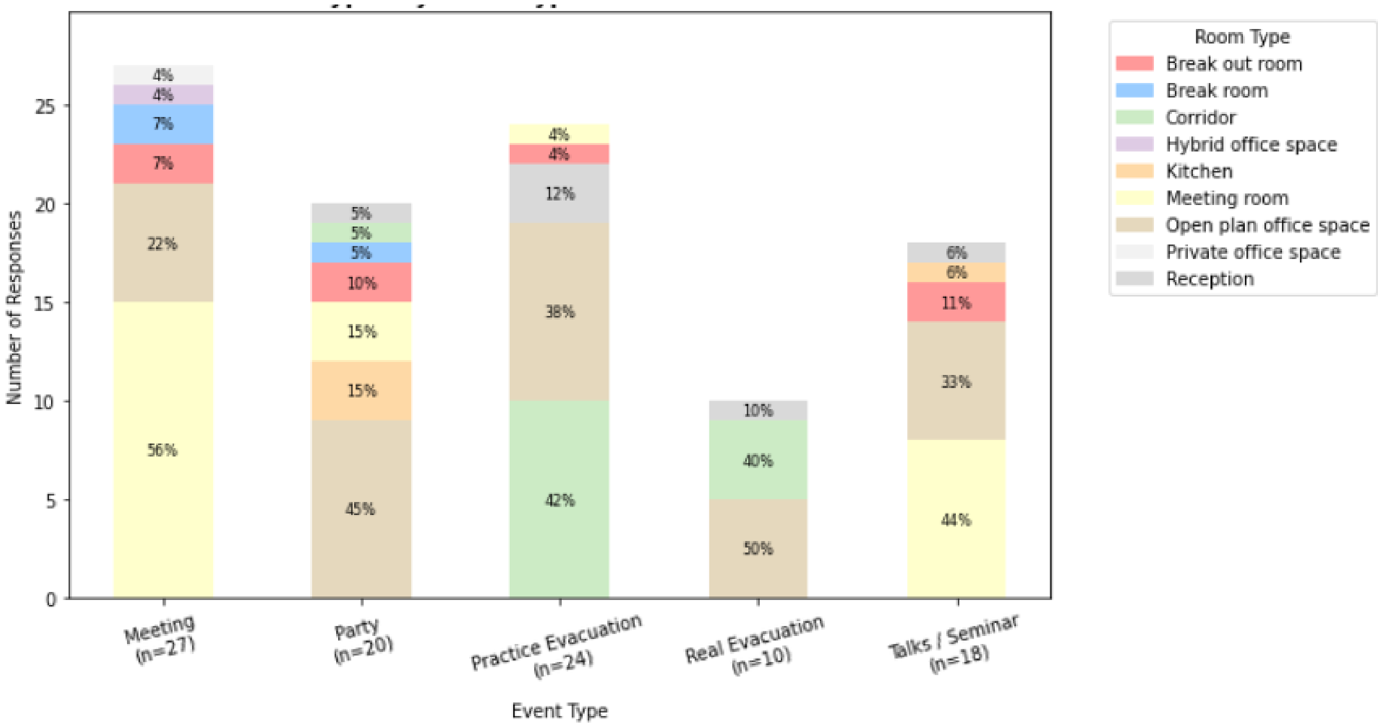


Figure 15: Stacked bar chart of reported room types for each event type.

Figure 16 shows the distribution of frequencies by event type, using the median value for time ranges. Due to the large variation in frequencies a logarithmic scale was used.

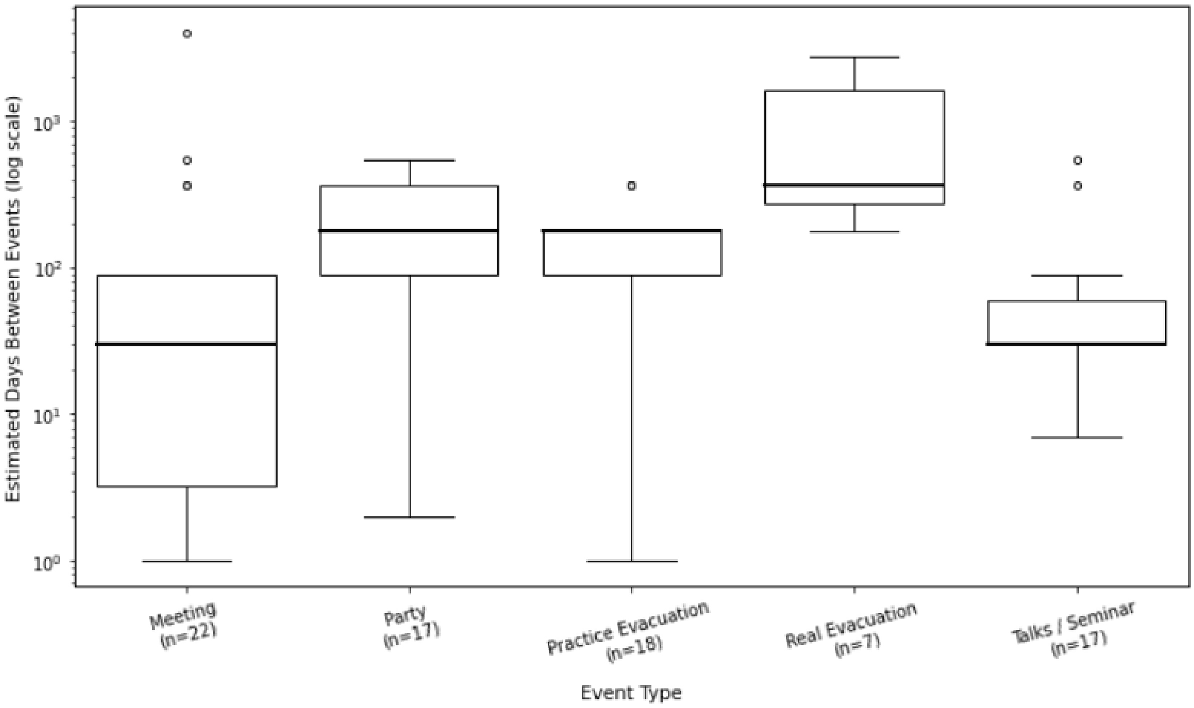


Figure 16: Reported frequency of event types.

4.3 Short term storage loading

Table 6 summarises the responses gathered relating to short term storage events. Comparable responses have been grouped together to simplify the table layout and reduce repetition. For the complete table see Table B1 in Appendix B. As participants made their own description of the items being stored, it meant that many unforeseen answers such as 'concrete block sculpture' were gathered. Additionally, it meant that unfortunately several responses such as 'materials' and 'boxes' were unable to be used, as there was not enough information regarding the item for a load to be assumed. Many responses also did not provide the area over which the storage acted and were also unable to be used. Additionally, some responses did not includes a frequency value, leading to the discrepancy between the number of responses for each item and the frequency value shown in the final column.

Table 6: Summary table of reported short term loading events (For full results see Table A1 in appendix)

Item(s) being stored	Number of Responses	Amount (Average)	How Stored	Floor Area of Storage(m^2) (Average)	How Often Does This Event Occur
Paper / Books	9	163.94	stacked on top of each other (3) stored on shelves or racks (3) piled in one location (3)	3.73	once every 1-2 months (4) once every 3-5 years (1) once every 3-6 months (1) one-off event (2) once per year (1)
IT Equipment	4	31.75	piled in one location (3) stored on shelves or racks (1)	1.15	once every 1-2 months (1) once every 6-12 months (1)
Misc Office	3	29.67	spread throughout the office (1) spread evenly across the floor (1) stacked on top of each other (1)	1.61	one-off event (2) once every 1-2 months (1)
Desks	3	10.5	spread evenly across the floor (3)	5.14	one-off event (1) once every 1-2 months (1)
Office Furniture	2	10.0	spread evenly across the floor (2)	3.0	one-off event (2)
Chairs	2	106.75	spread evenly across the floor (1) stacked on top of each other (1)	7.17	once every 1-2 months (2)
Photocopier	1	1.0	piled in one location (1)	0.42	
Water Bottles	1	500.0	stacked on top of each other (1)	2.67	once every 1-2 months (1)
Water Dispenser bottles	1	4.5	piled in one location (1)	1.0	once every 1-2 months (1)
Tower PC	1	150.5	stacked on top of each other (1)	3.75	once every 2 years (1)
Supermarket Delivery	1	4.5	stacked on top of each other (1)	1.0	weekly (1)
Monitors	1	4.5	in a row (1)	1.0	once every 3-5 years (1)
Concrete Block Sculpture	1	88.0	stacked up to (1) blocks high (1)	3.75	once every 2 years (1)

5.0 Discussion

This section interprets the survey results and assesses their ability to evaluate whether the findings support the studies mentioned in the literature review. Through analysis of respondent demographics, event-based data, and storage loading information, the study aims to draw comparisons, conclusions and recommendations to hopefully aid the understanding and analysis of extreme loading events in office environments.

5.1 Analysis of Respondent Demographics:

From Figure 10 it can be seen that most survey respondents work within the realm of engineering or technical work, with almost 60 percent of people choosing this option. This disproportionate representation is likely due to the method of survey distribution, as it was posted to the author's and supervisor's LinkedIn which predominantly connect with engineers. Although this may provide good insight into a technical office environment, this may influence the reliability of the data gathered, as values may be heavily skewed towards offices that often contain specialised equipment, leading to over representation of high intensity loading events especially regarding storage loads and their frequency. However, the technical background may lead to a better understanding of the aim of my study and therefore more meaningful responses.

Figure 11 shows how respondents' time spent in their current office varied greatly. Converting results to the median value of the time ranges selected, the mean duration in the current office was 4.1 years, matching data from the U.S. Bureau of Labour Statistics (2024) for the median number of years salaried workers had been with their current employer as of January 2022. This suggests a representative sample of the office industry was gathered. However, roughly 25% of respondents had under a year working in their current office. Although this may be representative of the current workforce dynamic, it raises concerns that these individuals may not have had sufficient opportunity to witness a high loading event. Additionally, with 60% of respondents having been in their office for under three years, much of the data reflects the post COVID working conditions. Current figures (OBeirne, 2025) show an average occupancy rate of 37.1% as of the last week of January 2025, this is drastically different to the 60-80% rates seen in pre pandemic workplaces. This could mean that as offices transition back to a pre covid way of working the values presented in this study may not be able to accurately map the extreme load cases.

To explore whether the duration of time spent in the current office correlated with the maximum load witnessed, Figure 17 plots the converted maximum witnessed density (in kN/m²) against respondents' years in the office using a log scale. Different shaped and coloured points have been used to represent different event types data. When values are in the same place, their size increases to show the frequency. More common responses have a higher weighting when lines of best fit are drawn in the corresponding colours. Surprisingly, respondents with under a year of experience report similar peak densities to those with over 10 years, while those in between reporting significantly lower values. This trend contradicts the expectation that those who had been in the office for longer periods of time would have higher exposure to extreme events. This result may reflect sample size limitations. However, it would mean the earlier concerns of having respondents with a short duration in their current office would not influence the max loads that they witnessed.

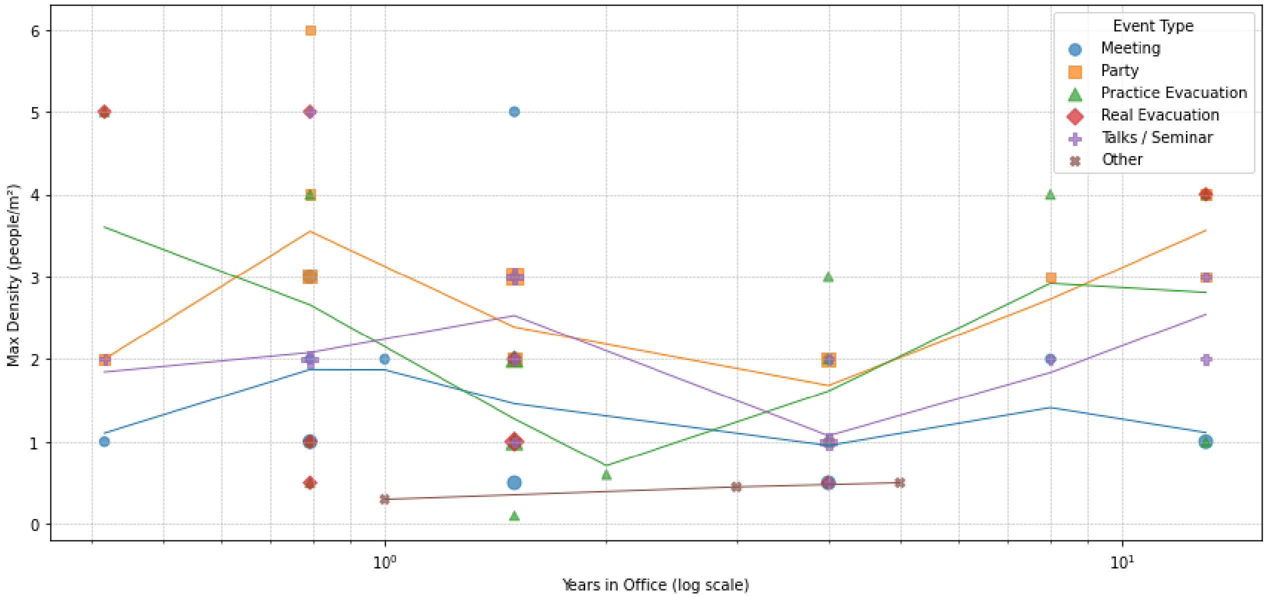


Figure 17: Log-scaled plot of maximum densities witnessed versus years in office, with marker size scaled to event frequency.

5.2 Interpreting Maximum Loads

To make results comparable to the values presented in the literature review, values of people/m² were converted to kN/m². This involves assuming a weight for a person, however this is more nuanced than it may first appear. Extreme loads happen with a relatively low number of people, as is highlighted by several respondents choosing the option of 1-5 people being at the maximum density. Therefore, the analysis cannot just assume the average weight of a person. This is because in large samples extreme values will cancel out and the mean value is very close to the expected. Conversely with small samples the values are influenced much more on a person-to-person basis. For example, a group of 120kg rugby players in a lift will have an extremely different load compared to a group of school children. Both events are likely to occur so the extreme value of the rugby players must be designed for and not the average of a rugby player and a child.

Unfortunately, data and research on the values and methods to analyse the weights of people to use for different samples sizes is extremely limited. Although Andam (1986) references a mean body weight of 67kg and a standard deviation of 11kg for a sample of 2000 office workers in his analysis, these values are likely outdated and not accurate, as weights have increased drastically since the time of his analysis (BBC News, 2010). The research found no method to calculate the average weight needed in design depending on the sample size in question using the standard deviation or any other parameters. Therefore, the study was unable to convert the number of people per meter squared reported into different weights depending on the average number of people reported at that density in the event, and a much more simplistic approach has been used.

The institute of Industrial and Systems Engineers (n.d.) state ‘Generally, design limits should be based upon a range from the 5th percentile (small) female to the 95th percentile (large) male values’. While this guidance doesn’t explicitly relate to body weights or floor loading it provides a strong precedent for adopting a percentile-based assumption. Therefore, this study has converted occupancy densities (people/m²) to load intensities (kN/m²), for all events, using the 95th percentile male body weight. From Figure 18 the value used in the study is 284lbs (129kg). Although this value is substantially higher than the 67kg Andam uses, I believe it more accurately represents the weights of a modern man.

It is assumed that gravity is equal to 10 (standard in many structural engineering calculations), and therefore that the load per person is 1.29 kN.

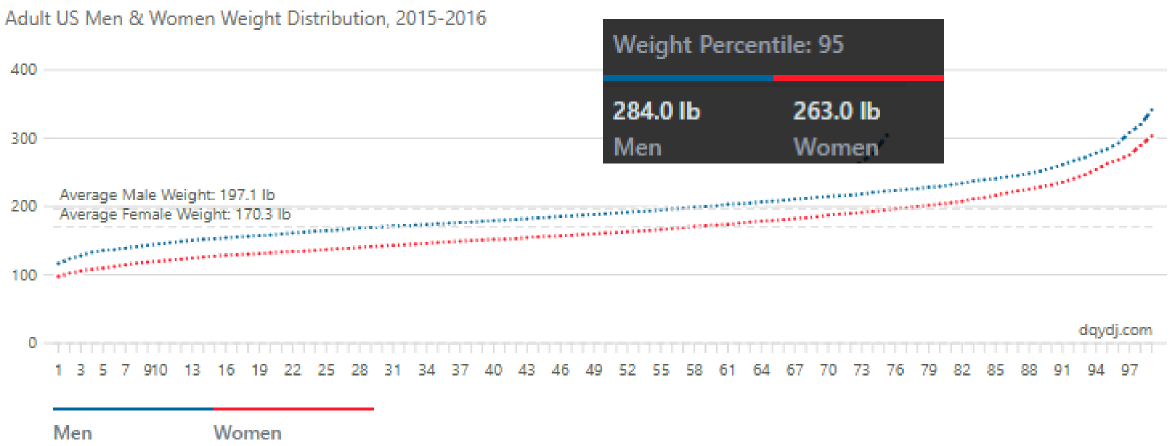


Figure 18: 95th percentile weight (PK, n.d.)

5.3 Event Based Data interpretation:

Respondents were given the option to choose population densities ranging from 0.2 to 7 people m². Figure 12 shows the maximum load witnessed was 7.74kN/m² (6 people/m²). Although this point was identified as an outlier it proves crucial in the mapping of the most extreme events. The median values across all events varied from 1.29 to 3.87 kN/m² (1 to 3 people m²), with the most common being 2.58 kN/m² (2 people m²). The range in upper and lower quartiles was 1.29 kN/m² to 5.16 kN/m² (1 to 4 people m²). If these values are compared to those presented in the literature review, insight into how accurate the previous work may be and how values have changed over time can be made. It can be seen that almost all the maximum values witnessed in surveys are below those of this study: Mitchell and Woodgate’s (cited in Alexander, 2002) value of 3.05kN/m², the BRS (cited in Alexander, 2002) value of 2.4 kN/m², and Dunham et al.(1952) value of 3.7 kN/m² all highlight the discrepancy between the maximum load witnessed. This possibly highlights how the personnel weights have changed drastically since the time of their surveys and how office environments have changed. This highlights how the models that rely on the outdated surveys need to have more up to date values that represent the loading of modern office environments. However, Mitchell and Woodgates (cited in Alexander, 2002) value of 2.4 kN/m² for crowd loading during a fire event maps fairly consistently with the median value of practice evacuations of 2.58 kN/m². This could show that the analysis that they conducted on these extreme events focused more on the average extreme event rather than the most extreme event.

This shows how many of the design codes and prescribed values could also underestimate the true extreme events, specifying loads lower than those witnessed in this study. The BCC's (1925) value of 4.8 kN/m² and the BCO value of 7.5 kN/m² are all lower than the maximum load witnessed in this study. This highlights that the models used to prescribe these values need to have strong justification with empirical data.

Although parties show a higher median load, their interquartile range is much narrower than many other event types, suggesting higher average occupancy but fewer extreme spikes. Contrastingly, evacuation scenarios show much greater variability with higher and lower extreme values. This shows that the potential most extreme peaks are likely to occur during this event type. This supports the view, that is backed by the literature in this study, that low frequency high intensity events are most critical for design. However, this reported data relies on user perception and may be over or underestimated. For example, values could be affected by psychological distortion during evacuation events where crowding feels more intense due to urgency and heightened awareness.

When the results of maximum density are instead plotted against room type, as in Figure 13, significant variability is apparent. This supports the idea that room-based modelling offers a more accurate reflection of the extreme loading likely to occur than event-based categorisation. This contrasts with the thoughts of Allen (1958) who believed that the use of the space was most important.

Figure 14 demonstrates that peak densities tend to be spatially localised, as median values for the number of people at the max density is less than the total amount of people at the event. In the responses gathered several results for both categories had chosen the maximum possible value, the author believes that these results to be misinformed inputs. However, they were not redacting and had little effect on the conclusions that could be drawn, showing only a smaller proportion of people at the max density. These findings support the models proposed that represent the extreme loads as occurring in small clusters rather than across the whole floor such as the work of McGuire and Cornell (1974). Once again, this figure shows the importance of understanding the short duration events, as practice evacuations show that although the median value may be under 100, the most extreme case saw values above 500.

The population of people at meetings seems to be significantly less than that of any other event, this could show that these are likely to be complete office wide events, that can have extreme amounts of people gathered at the same time, whereas meetings are normally conducted in smaller groups. Although talks and real evacuations have a similar total population, the amount of people at the maximum density is far greater during the evacuation. This could show that planned events, such as meetings, have a safer max density that more of the group stay at, whereas the evacuations are more prone to much greater extreme events such as bottlenecks at doorways, meaning that a small proportion of the total population are at much denser extremes. It underscores the need to consider crowd movement and bottleneck effects in structural loading models, which is something that has not been widely integrated.

However this may be an oversimplification understanding of the problem, and the reason why the extreme events have low amounts of people at the maximum density could be because there is no ability to see everyone who is at this maximum density. Respondents may know for example that their whole building was evacuated and therefore that it affected a large amount of people but they were only able to see those on their route out of the building. Conversely, at a talk, respondents are likely to be able to see everyone at the event as they will be in the same room and therefore can see how any people are at the maximum densities. These results bring into question the assumption that full occupancy always results in maximum loading. In practice, peak structural demand may arise from the smaller grouping of people within the event.

The information portrayed in Figure 15 shows that the type of event occurring has a huge impact on the room in which the event is likely to be witnessed. For example, unsurprisingly 56% of people chose a meeting room as the room type where a meeting had occurred, and only 4% of respondents reported a practice evacuation in a meeting room. This shows how different areas, even within the same office are likely to experience extremely different worst-case loading. This correlates nicely with the BCC (1925) and shows that designing higher loading in areas where the events are likely to occur instead of everywhere is a much more efficient approach. The most reported room type in which extreme events occur was open plan office spaces, meeting rooms, and corridors. This could show that these areas are the ones that need most stringent, in-depth design as they are likely places for any of the extreme event types to occur. The finding that 15% of responses for parties occurred in a kitchen is interesting as it shows they are an important location for extreme events. This contradicts the findings of Andam (1986) who decided to omit kitchens from his survey thinking that they were unimportant.

Another key inference that can be gathered from figure 15 is that, due to receiving far fewer responses on real evacuations compared with any other event type, there is an inherent challenge that collecting empirical data on rare but potentially high impact events poses. This serves as evidence that the combination of models and empirical values are needed together as both have their drawbacks.

The respondents were given the opportunity to choose multiple room types for each scenario. Many respondents did, and this could show that sometimes events occur in different areas at the same time or that they progress through a building. This would be expected in an event such as an evacuation, people leave their office area, into a corridor, and through the reception. This flow through the building causes sequential peak loads rather than isolated crowding zones. Existing models do not seem to be able to map this as they have no ability to trace the load through the building.

Another key factor of extreme loading events is the frequency of their occurrence as Figure 16 highlights. Although the events seem to have long lower quartile whiskers, this is due to the logarithmic scale used to map the frequency. Therefore, all events seemed to have relatively low interquartile ranges suggesting that these events have consistent scheduling across offices. The median values for meetings and talks sit almost exactly on a month showing that these events are likely scheduled for this interval of time widely across office types. However, on the other end of the spectrum, real evacuations median value is roughly 500 days with a much larger interquartile range. This shows that this event type likely happens less frequently and with random distribution.

5.4 Analysis of storage loads

To convert the information shown in Table 6 to more meaningful values, data from the ranges used in the ‘amount’ column were converted into distinct values using the median. Weights were then assumed for each item and a value of weight per meter squared was calculated using the floor area of storage. There was one extreme outlier giving a value of 1296kg per meter squared, coming from what was described as ‘office supply reorganising’, however the author believes this was due to the respondent answering the floor area for each item rather than as a collective.

Table 6 provides an insight into how varied office-based storage practices can be in both magnitude and frequency. The table shows how varied the extreme event loading can be, with multiple responses for storage areas of less than 0.25 m^2 and above 25 m^2. The results also highlight the range of storage methods reported, which adds further complexity to the analysis of their loading. Additionally, items that were commonly reported such as paper storage can be seen to have a surprisingly high load density. This shows how the storage load, which is often overlooked in studies, may be the most important factor as the frequency of some of the extreme events far outweigh those relating to population densities.

The need to assume weights for items leads to a high degree of uncertainty in the values that they produce. Therefore, the conclusions presented here regarding the extreme loading due to short term storage must be approached with caution. Although the study does compare values to those in the literature, it is more important to notice the trends that the data shows over the exact values. This highlights how difficult the collection of data regarding these parameters is and that more studies into these load cases needs to be conducted.

To see if there was a correlation between the densities calculated and the frequency of their occurrence. The values of weight/m^2 were plotted against the median value of the frequency ranges chosen as can be seen in Figure 19. Where events were chosen to be ‘one off events’, a frequency of every 20 years was chosen, as this was the maximum duration that any of the respondents had been in the office. Unfortunately, much of the data received could not be used in this figure as many responses did not give the frequency of the event. This meant that although most people gave multiple examples of storage loading, there were still only 21 values that could be used. However, the graph still shows a slight positive gradient indicating that less frequent events are the ones likely to cause the maximum loads. This is also reinforced by the highest value reported being a one-off event. The graph however shows a wide distribution of points that do not fit closely to the best fit line and therefore that this correlation could be rather weak. However, if a line more heavily focused on the most extreme values rather than the median was plotted, the author believes it would show a much stronger positive correlation. This suggests that in reality more extreme loads do occur less frequently.

It can also be seen that the most extreme loading due to the storage of items was roughly 400kg/m^2 (4.0kN/ m^2). If this is compared to the most extreme values presented in the literature, as was done for personnel loads, it can be seen that a value of value of 3.7 kN/m^2 presented by Dunham et al. (1952) is quite close to the witnessed value. However, Mitchell and Woodgate’s (cited in Alexander, 2002) value of 3.05kN/m^2 and the BRS (cited in Alexander, 2002) value of 2.4 kN/m^2, are still much smaller than the values seen in this study. This value of the extreme event can also be compared against the prescribed values such as the BCO's value of 7.5 kN/m² which is drastically higher than the maximum loads witnessed in this study due to storage events. However, the value of 4.8kn/m² presented by the BCC (1925) is much closer to this survey's results. Although these studies focused on the population densities the comparison shows how this study's findings relate to the data and prescribed values. Another meaningful comparison can be drawn between Allen's (1958) analysis of the Canadian national building code, their equivalent value of 15.6kN/m² to represent an extreme point load is drastically higher than the presented value. This could show that office uses have changed over time and these extreme values of storage load are no longer witnessed in a modern environment where most paper storage events have reduced due to the use of computers.

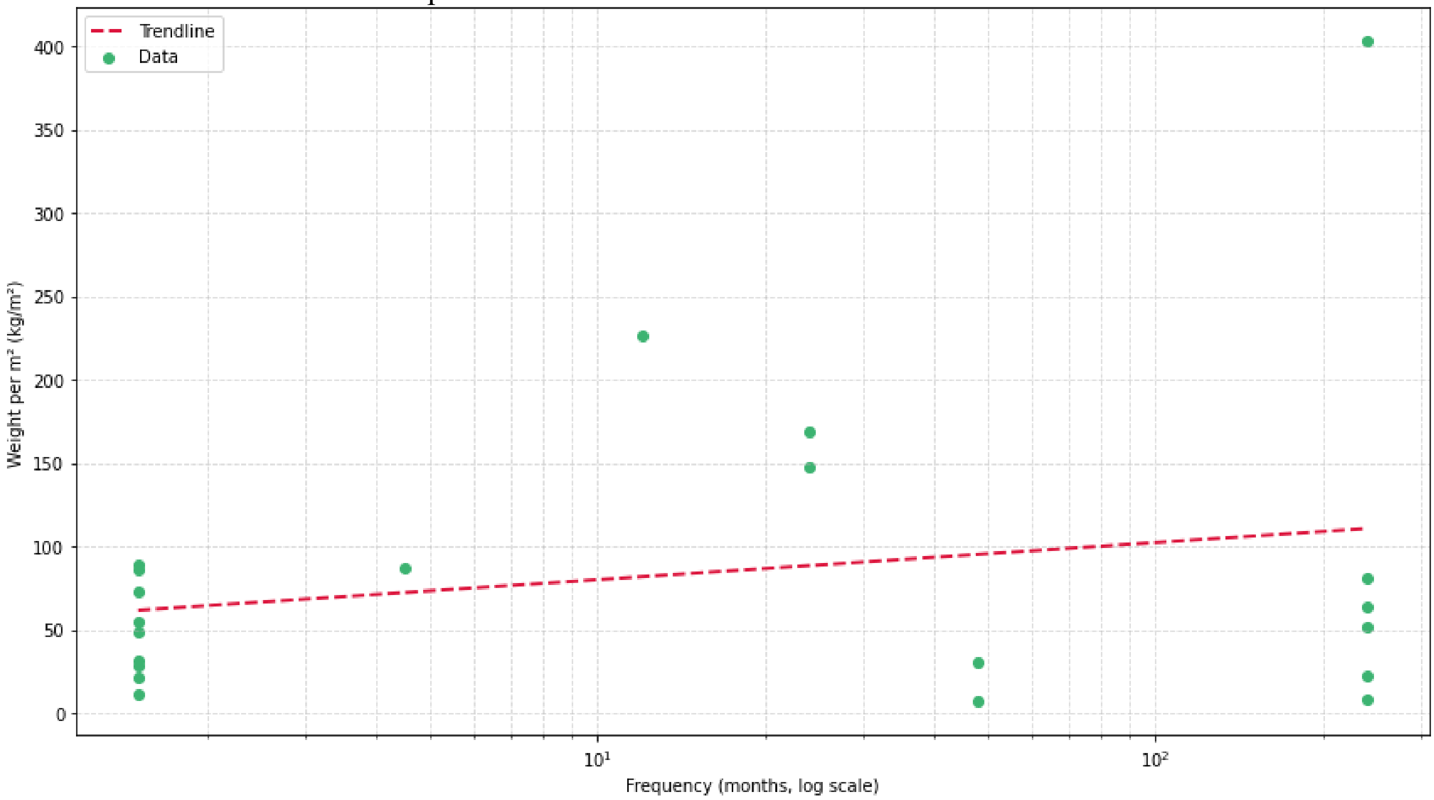


Figure 19: Weight per m^2 against frequency using a logarithmic scale

6.0 Conclusion

This dissertation set out to analyse peak loading conditions in office environments, with a focus on short term high occupancy events and localised storage to address the gap between theoretical and empirical approaches to floor loading. The central aim was to contribute to more realistic, efficient, and adaptable structural design practices by providing updated and contextually relevant empirical data. Through a comprehensive analysis of existing literature and the gathering of first-hand data via a survey, prevalent load values and underlying assumptions in traditional, statistical, and probabilistic models were identified and assessed.

This survey gathered real-world data on maximum population densities witnessed during different event types, their frequency, associated total populations, and the room types in which they occurred. Additionally, this survey gathered data on storage loading, focusing on the distribution and frequency of heavy storage events within typical office spaces.

The comparison and analysis of the results lead to many key findings:

- Historical survey data frequently underrepresents modern extreme event population densities, possibly reflecting the increase in weight of the population, or the changing office environment.
- Existing design recommendations for extreme loads also seem to generally underrepresent the extreme values presented in this study.
- Short term storage loading, which is overlooked in almost all models and surveys, demonstrated a higher maximum weight per unit area than those found in previous surveys relating to population densities. This highlights the important gap in both traditional and probabilistic analysis.
- Peak loading events were often found to be highly localised rather than uniformly distributed, which agrees with most modern modelling approaches considering spatial distributions.
- Widely accepted stochastic models of extreme loading may oversimplify real-world scenarios, as findings indicate that significant loading events often occur at regular predictable intervals.

The analysis and data gathered in this dissertation aim to contribute to a more accurate understanding of extreme loading events in office environments, and therefore efficient structural design. The findings provide empirical values and trends that can support the development of more robust analytical methods. These methods can address the limitations of existing models and assumptions identified throughout this study.

7.0 Recommended Further Research

One of the key limitations of this study was the small sample size. Despite the questionnaire having over 700 views there were only 25 meaningful responses. This small sample size reduced the statistical reliability and meant values may not have been representative of a true population. A larger sample size would help identify clearer trends and give more accurate results. In future studies, this could be achieved by keeping the survey open for a longer period or by offering incentives to increase participation.

Another limitation was the format of the questionnaire responses. Although the dropdown menus with numeric ranges were used to simplify the process and encourage participation, they inherently reduce the accuracy of the data. Exact numerical entries would provide more precise and representative results, especially for questions where large ranges were used such as the number of people witnessed.

Additionally, the reliance on respondents' memory likely made inaccuracies in their responses. The survey focused on the maximum loading events that they had ever witnessed in their office and therefore the events that they were describing and giving data for could have been several years ago. This introduces significant potential for memory bias or simply a lack of memory regarding true high loading events. A more accurate approach could be to conduct a longitudinal study whereby participants record data shortly after the peak loading event has occurred.

A key parameter used in this study was the weight used to convert from people/m^2 to kN/m^2 there was little research regarding how to accurately assume a weight that accounts for small groups of people. Therefore, measurement and analyses of such events is crucial to ensure accuracy of findings similar to this dissertation.

Another potential avenue of research stemming from this study is the modelling of the dynamic effect that occurs during these high population density events where frantic sporadic movements of large amounts of people are likely to occur, such as during a fire evacuation.

Future research should use the values and trends presented from the data in this study to serve as a foundation for further more-detailed analysis that could potentially change the way design and modelling of extreme loads is undertaken.

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